

CHAPTER 3: CLASSIFICATION OF ELEMENTS & PERIODICITY (JEE FOCUS)

Topics Covered:

- Modern periodic table
- Periodic trends
- Electronic configuration
- Atomic radius
- Ionization enthalpy
- Electron gain enthalpy
- Electronegativity

Core Concepts:

- Periodic law → properties periodic with atomic number
- Groups → same valence electrons
- Periods → same principal shell
- Atomic radius ↓ across, ↑ down
- Ionization enthalpy ↑ across, ↓ down
- Electron gain enthalpy → more negative across
- Electronegativity ↑ across, ↓ down
- Effective nuclear charge ↑ across
- Shielding effect ↑ down
- d-block → poor shielding
- Anions > atoms > cations (size)

Key Formulas:

- $Z_{\text{eff}} = Z - \text{shielding}$
- No fixed formula → trend-based reasoning
- Ionization enthalpy $\propto Z_{\text{eff}} / \text{size}$

Important Patterns:

- Across → size ↓, IE ↑, EN ↑
- Down → size ↑, IE ↓
- Exception → B < Be, O < N (IE)
- EA exception → Cl > F
- Cations smaller, anions larger
- Isoelectronic → higher Z → smaller size

Common Mistakes:

- Ignoring exceptions (B, O cases)
- Confusing IE vs EA
- Wrong trend direction
- Ignoring shielding
- Not using isoelectronic logic

Examples (Important Cases):

- Compare size: O^{2-} , F^- , Na^+
- IE trend in 2nd period

- EA: Cl vs F
- Order of atomic radius
- Electronegativity comparison

Exam Tips:

- Always apply trend, not memory
- Check exceptions carefully
- Use elimination in MCQs
- Focus on isoelectronic species
- Visualize periodic table mentally